

Spatial Analysis of Fishing Effort within Marine Protected Areas in the Mexican EEZ

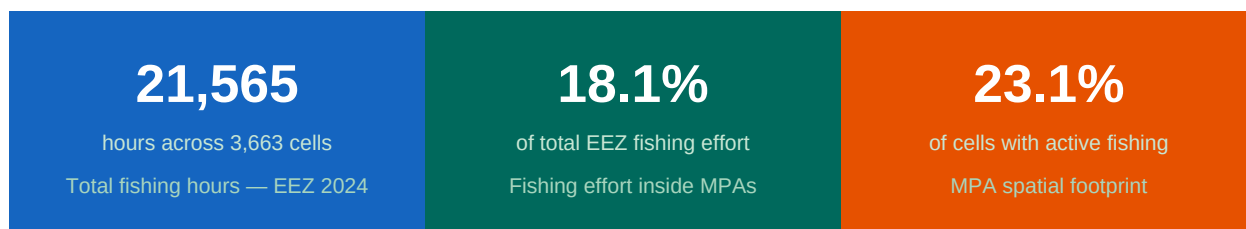
Global Fishing Watch · CONANP · QGIS — Fishing Pressure Assessment Inside
Mexican MPAs (2024)

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Tools: Global Fishing Watch · QGIS 3.40 · R · CONANP · MarineRegions.org

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1. Introduction

According to the National Commission of Natural Protected Areas (CONANP), Natural Protected Areas (*Áreas Naturales Protegidas*, ANP) are defined as territories under national jurisdiction where original ecosystems have not been significantly altered by human activity, or whose ecological functions require preservation or restoration. These areas are subject to the legal regime established in Mexican environmental law and other applicable provisions.

As of 2024, Mexico has approximately 232 federal protected areas, categorized into Biosphere Reserves, National Parks, Flora and Fauna Protection Areas, Natural Resource Protection Areas, Natural Monuments, Sanctuaries, and Voluntarily Conserved Areas. Out of the total, around 40 are marine protected areas (MPAs), belonging to four categories: Biosphere Reserves, National Parks, Flora and Fauna Protection Areas, and Sanctuaries. Among them, National Parks are the most numerous (17), followed by Biosphere Reserves (16) (CONANP, 2024).

MPA Categories

- **Biosphere Reserves:** Ecologically significant areas with representative, largely unaltered ecosystems. Core zones allow only preservation, scientific research, and environmental education; buffer zones permit regulated sustainable use.
- **National Parks:** Zones characterized by scenic beauty, scientific value, recreational or historic significance, or rich biodiversity. Conservation, tourism, research, and education are permitted under protective management.
- **Flora and Fauna Protection Areas:** Habitats whose conservation is essential for the survival of specific plant and animal species. Regulated sustainable use, research, and propagation are allowed.
- **Sanctuaries:** Protect specific microhabitats or geographically restricted ecosystems supporting unique or vulnerable flora and fauna. Only scientific, educational, and recreational activities compatible with conservation goals are allowed.

All MPAs lie within Mexican waters and the country's Exclusive Economic Zone (EEZ), which extends up to 200 nautical miles (370.4 km) offshore. Within this zone, Mexico holds sovereign rights for the exploration, exploitation, conservation, and management of marine natural resources, regulated under Articles 46–56 of the Federal Law of the Sea (CONABIO–SIMAR, 2025).

The Mexican EEZ is the primary region where national fisheries operate. However, compliance with MPA regulations is not always guaranteed, and many protected areas experience significant fishing pressure that may undermine their ecological integrity and management effectiveness. For this reason, the present study aims to analyze the fishing effort within MPAs in the Mexican EEZ, identify which MPAs are most affected, and provide insights to support evidence-based conservation strategies.

2. Objectives

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- Quantify total and mean fishing effort within the Mexican EEZ for 2024 using Global Fishing Watch data.
 - Assess the proportion of fishing effort occurring inside Marine Protected Areas.
 - Identify MPAs with the highest total and average fishing effort per cell.
 - Map the spatial distribution of fishing pressure relative to MPA boundaries.
 - Provide evidence-based recommendations to improve MPA management and enforcement.

3. Methods

3.1 Data Sources and Preparation

Fishing effort data for 2024 were obtained from Global Fishing Watch in raster format, representing the total number of fishing hours per 0.1-degree grid cell within the Mexican EEZ. The raster dataset was pre-clipped to the EEZ boundary to ensure spatial consistency. MPA shapefiles were retrieved from CONANP and intersected with the EEZ layer. All spatial layers were projected in WGS 84 (EPSG:4326) and processed using QGIS 3.40. Additional spatial data (including coastlines and EEZ boundaries) were sourced from MarineRegions.org.

3.2 Analytical Workflow

The analysis followed six sequential steps:

- **Effort characterization in the EEZ:** Basic statistics were calculated for all non-NaN cells in the effort raster, including sum, mean, standard deviation, maximum, and coefficient of variation (CV).
- **MPA overlay:** The raster was clipped using the MPA polygon layer to isolate fishing effort within protected areas.
- **Statistical comparison:** The same summary statistics were computed for raster values within MPAs.
- **Percentage calculations:** The proportion of valid effort cells and total fishing effort within MPAs was computed relative to the entire EEZ.
- **Distribution analysis:** A histogram of fishing effort per cell was generated using a log-scale y-axis to better visualize skewness and outliers.
- **Map creation:** A thematic map displaying fishing effort intensity and MPA overlap was produced in QGIS using five effort categories: Low (0–50 h), Moderate (50–150 h), High (150–300 h), Very High (300–400 h), and Extreme (>400 h).

4. Results

4.1 Fishing Effort in the Mexican EEZ

A spatial analysis of fishing effort within the Mexican EEZ during 2024 revealed a total of 21,564.7 fishing hours distributed across 3,663 valid grid cells. The average effort per cell was 5.89 hours, with values ranging from 0 to 441.4 hours. The standard deviation was 14.99 hours, yielding a coefficient of variation of 254.6%, indicating a highly uneven distribution of fishing activity with intense concentrations in specific coastal and offshore regions.

Statistic	EEZ	MPAs
Valid cells (n)	3,663	847
Total effort (hours)	21,564.7	3,915.1
Mean effort per cell (h)	5.89	4.62
Standard deviation	14.99	8.77
Coefficient of variation	254.6%	189.8%
Max effort (hours)	441.4	120.4

Table 1. Summary statistics of fishing effort in the Mexican EEZ and Marine Protected Areas (2024).

Within the MPA network, 847 grid cells recorded valid effort values, accounting for 3,915.1 hours, representing 18.1% of total EEZ fishing effort. MPAs comprised 23.1% of the spatial extent with active fishing. The average effort within MPAs was 4.62 h/cell (SD = 8.77; CV = 189.8%). These results suggest a partial reduction in pressure due to protective measures, though the presence of moderate to high effort values within certain MPAs raises concerns regarding compliance, enforcement, and MPA design.

4.2 Effort Distribution

The log-scaled histogram (Figure 1) highlights a strong positive skew: the vast majority of grid cells register fewer than 10 fishing hours, while a small number exceed 100 hours. This reflects the presence of distinct fishing hotspots within the EEZ.

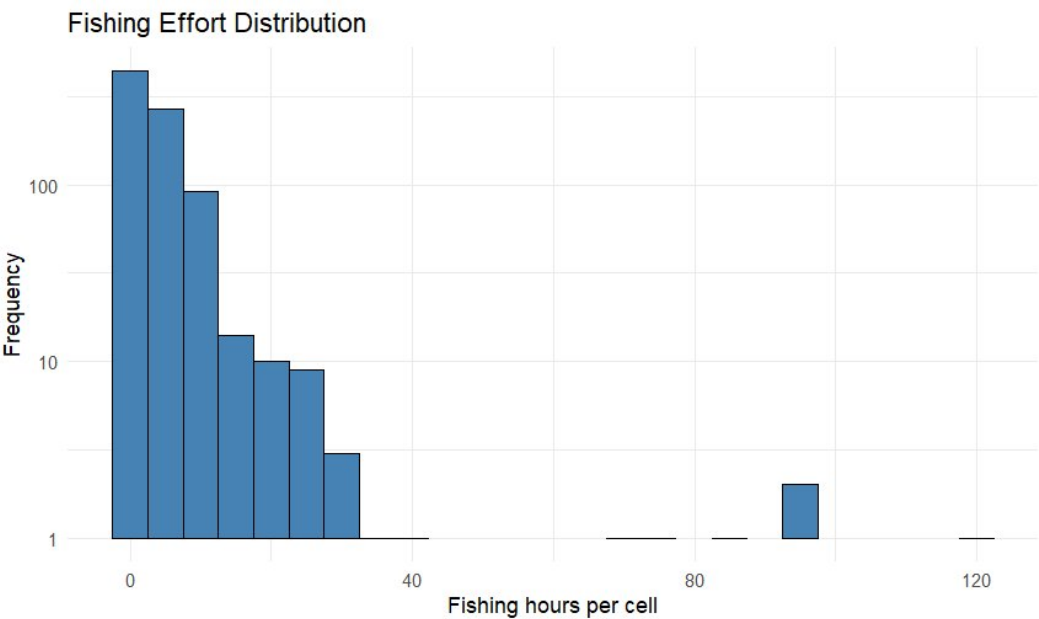


Figure 1. Distribution of fishing effort per cell in the Mexican EEZ (2024), displayed on a log scale. Most cells show low effort (<10 h), with a small number of extreme outliers exceeding 100 h. Source: Global Fishing Watch. Analysis: Israel Vallejo Muñoz.

4.3 Spatial Distribution and MPA Overlap

The thematic map (Figure 2) confirms spatial heterogeneity in fishing effort across the Mexican EEZ and clearly displays areas of overlap between high fishing effort and MPA boundaries, particularly in the Pacific region. Low-effort cells (0–50 h) are widespread, while moderate to extreme effort is concentrated along the Baja California Peninsula and select offshore areas.

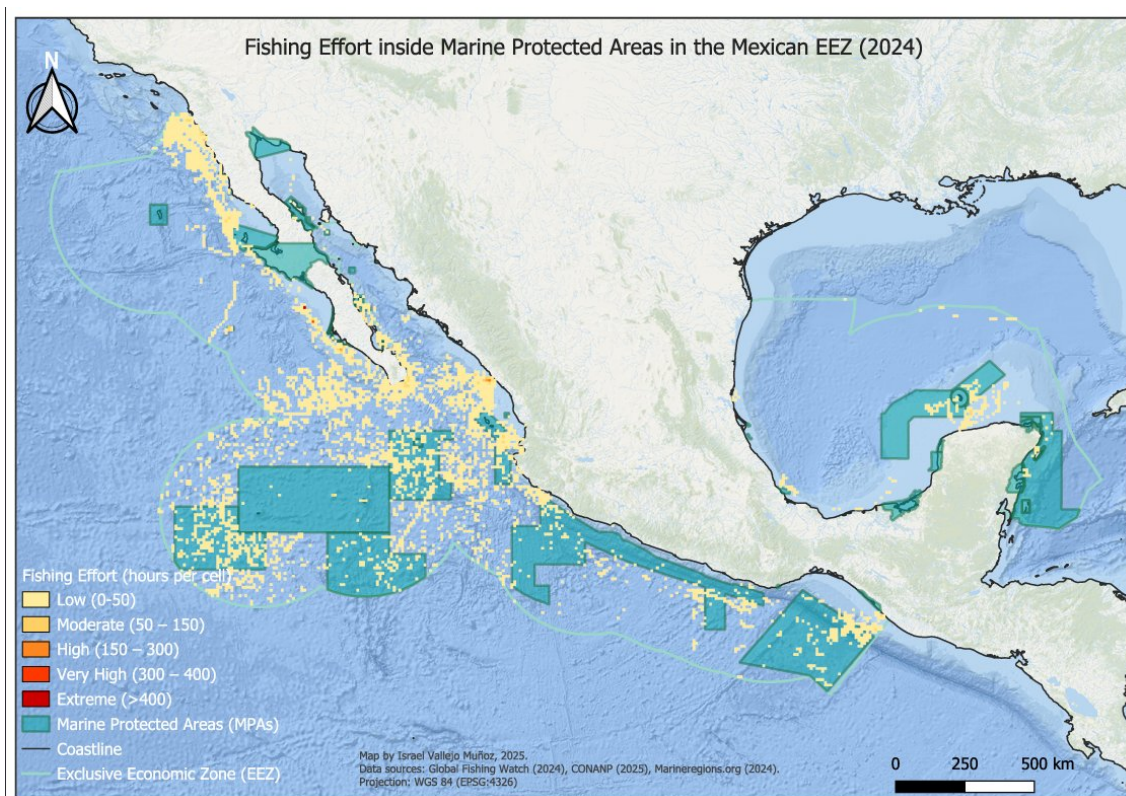


Figure 2. Spatial distribution of fishing effort within Marine Protected Areas in the Mexican EEZ (2024). Five effort categories are shown from low (yellow) to extreme (dark red). MPAs shown in teal. Source: Global Fishing Watch / CONANP / MarineRegions.org. Map: Israel Vallejo Muñoz.

4.4 Top MPAs by Total Fishing Effort

The MPA with the highest total fishing effort in 2024 was *Pacífico Mexicano Profundo*, registering over 2,300 cumulative fishing hours, followed by *Islas del Pacífico de la Península de Baja California* and *Arrecifes del Golfo de México-Sur* (Figure 3). These MPAs are generally large in size, which partly explains the high total effort, but may also reflect their accessibility and biological productivity.

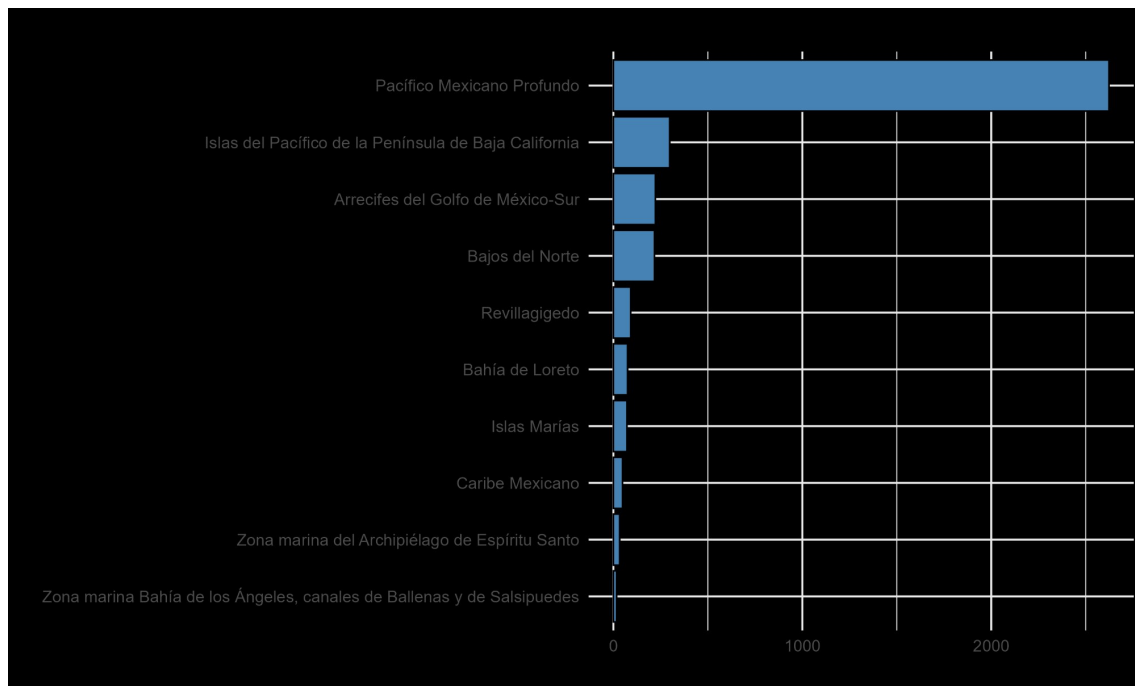


Figure 3. Top 10 Marine Protected Areas with the highest total fishing effort in the Mexican EEZ (2024), in cumulative fishing hours.

Source: Global Fishing Watch / CONANP. Analysis: Israel Vallejo Muñoz.

4.5 Top MPAs by Mean Effort per Cell

When analyzing average effort per cell (a metric that reflects local fishing intensity regardless of MPA size), a different pattern emerged. *Islas del Pacífico de la Península de Baja California* again topped the list, but smaller MPAs such as *Bajos del Norte*, *Bahía de los Ángeles*, and *Revillagigedo* also ranked highly (Figure 4). These results indicate that while some MPAs experience widespread pressure due to their size, others face localized but intense fishing activity that could threaten key ecological functions.

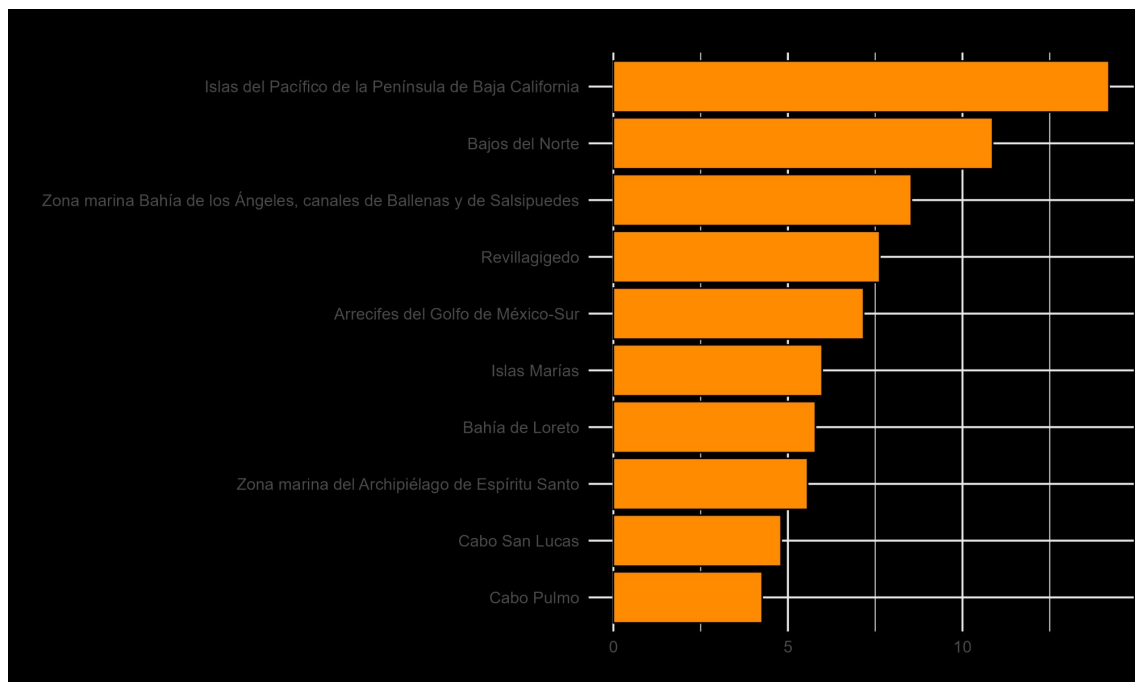


Figure 4. Top 10 Marine Protected Areas with the highest mean fishing effort per cell in the Mexican EEZ (2024), in average fishing hours per 0.1° grid cell. Source: Global Fishing Watch / CONANP. Analysis: Israel Vallejo Muñoz.

5. Discussion

The analysis of fishing effort within Mexico's MPAs revealed clear patterns of spatial concentration, regulatory gaps, and varying vulnerability levels among protected areas. While MPAs are designed to conserve biodiversity and maintain ecosystem functions, this study confirms that fishing activity remains a persistent pressure within many of them.

In 2024, approximately 18.1% of total fishing effort in the EEZ occurred within MPAs, despite these areas comprising 23.1% of the spatial footprint of fishing activity. This indicates a lower-than-expected concentration of effort inside MPAs, suggesting some protective effect, but not full exclusion of fishing activities. The high coefficients of variation both inside MPAs (189.8%) and across the EEZ (254.6%) confirm that fishing effort is highly clustered, with certain zones experiencing disproportionately high use.

Key Finding: While MPAs cover 23.1% of the fishing spatial footprint, they account for only 18.1% of total effort, suggesting partial but incomplete protection. Critically, several high-effort hotspots overlap directly with federally protected MPA boundaries, particularly *Pacífico Mexicano Profundo*, *Islas del Pacífico de la Península de Baja California*, and the newly designated *Bajos del Norte*.

The ranking of MPAs by total and average effort reveals two complementary patterns. *Pacífico Mexicano Profundo*, with over 43 million hectares of protected seafloor, experiences the highest total fishing effort. This enormous offshore biosphere reserve safeguards deep-sea ecosystems starting at 800 m depth, including hydrothermal vents, abyssal plains, submarine ridges, seamounts, and ocean trenches. While its vastness partly explains the accumulated effort, the ecological sensitivity and limited knowledge of its biodiversity call for urgent monitoring and regulation.

In contrast, *Islas del Pacífico de la Península de Baja California* (which includes ecologically rich archipelagos like Bahía Magdalena, Isla Cedros, and Archipiélago San Benito) showed both high total and average fishing effort per cell. These islands host remarkable diversity of marine mammals, seabirds, and endemic species. Their accessibility and productive coastal environments make them particularly vulnerable to sustained fishing pressure.

The recently designated *Bajos del Norte* Flora and Fauna Protection Area, located in the Campeche Bank of the Gulf of Mexico, also ranked among the most intensely fished. Although established in 2024, its core zone (344,467 ha) and buffer zone (959,647 ha) already show significant fishing activity, highlighting the need to implement immediate management measures and monitor compliance from the early stages of protection.

These findings emphasize that size alone does not guarantee protection. Large MPAs accumulate more effort due to their extent, while smaller MPAs can suffer from high localized pressure threatening ecological integrity. A one-size-fits-all approach is therefore insufficient; differentiated management strategies tailored to each MPA's size, location, and vulnerability are essential.

6. Conclusions and Recommendations

This study provides a spatially explicit assessment of fishing pressure within MPAs in the Mexican EEZ, revealing critical mismatches between legal protection and actual fishing activity. While MPAs show slightly lower average effort than the broader EEZ, significant fishing persists within their boundaries. Several MPAs, particularly *Pacífico Mexicano Profundo*, *Islas del Pacífico de la Península de Baja California*, and *Bajos del Norte*, exhibit disproportionately high total or average effort, warranting urgent attention.

The effectiveness of marine conservation depends not only on legal MPA designation, but also on consistent enforcement, ongoing monitoring, and adaptive governance. The following management strategies are recommended:

- **Enhanced surveillance** in MPAs with high total effort (e.g., *Pacífico Mexicano Profundo*), especially in offshore or hard-to-monitor areas.
- **Targeted enforcement** (including seasonal closures and participatory monitoring) in MPAs with high average effort per cell (e.g., *Bajos del Norte*, *Islas del Pacífico*), where localized fishing intensity poses significant ecological risks.
- **Early enforcement protocols**, community outreach, and ecological baseline studies in newly designated MPAs to ensure protection begins effectively from the outset.
- **Integration of satellite-derived fishing effort data** with in situ patrol reports and biodiversity monitoring to support adaptive, evidence-based conservation and to better prioritize management actions.

By aligning spatial data, enforcement tools, and ecological priorities, marine spatial planning in Mexico can more effectively meet its conservation goals and ensure the long-term sustainability of its protected marine ecosystems.

7. References

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